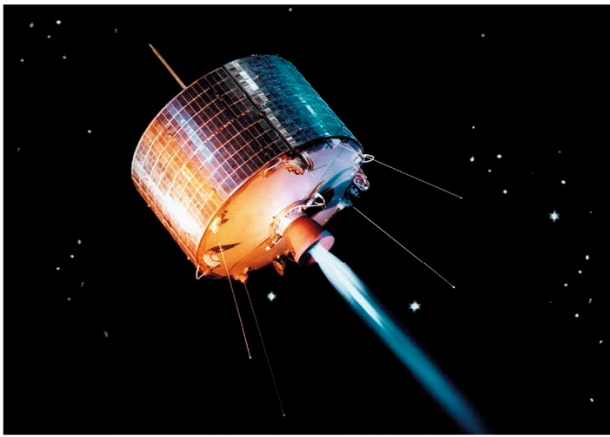


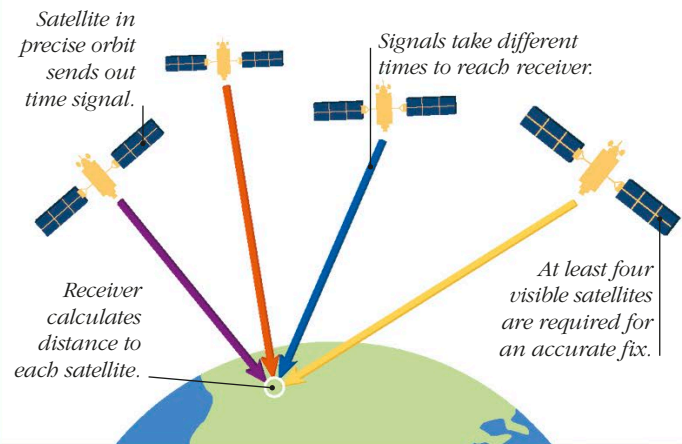


SYNCOM

Science-fiction author Arthur C. Clarke realized in 1945 that satellites in high “geostationary” orbits would be able to relay global communication signals. They would hang in a fixed position over the equator, so antennae could bounce signals over a large area via a satellite without having to track it. The first satellite to put Clarke’s idea into practice was Syncom 3, which relayed live television coverage of the 1964 Tokyo Olympics.

HOW GPS WORKS

The Global Positioning System (GPS) and similar navigation systems rely on networks of satellites in precisely known orbits. Each satellite carries a super-accurate atomic clock and transmits a time signal continuously. A computerized receiver measures how long it takes signals from different satellites to arrive, and uses that information to calculate the distance between itself and each of the satellites. Together, the measurements pinpoint the receiver’s exact location.



CALLS TO ORBIT

Most international telephone calls travel under the sea in fiber-optic cables, avoiding the time delays introduced when relaying signals over a long distance through ground stations via comsats in high orbit. But what if you are in a remote location with no link to a cell or landline network? Satellite telephones send signals directly to a satellite in low orbit, enabling communication from almost everywhere. They were originally designed for ships at sea, by the Inmarsat organization in the 1980s.

WOW!

GPS navigation receivers can pinpoint your location to within 16ft (5m)—Europe’s Galileo system can do it to the nearest centimeter!

► SATELLITE HANDSET

Today, satellite phones are widely used by explorers and rescuers in isolated regions.

